

Grading Criteria for Capstone Project

The Capstone Project grade is based on the implementation, analysis, presentation, and a written abstract.

Assumed contribution to final grade: **40%** (adjust as needed)

- ▶ The project involves developing a model, implementing core and additional features, analyzing results, and presenting findings.
- ▶ Deliverables: Presentation and a scientific abstract.

Capstone: Project Implementation and Technical Merit (40% of Capstone Grade)

▶ **Core Model Implementation (15%)**

- ▶ Accuracy and correctness of the fundamental model.
- ▶ Demonstrates understanding of core concepts.

▶ **Additional Features Implementation (15%)**

- ▶ Successful and meaningful implementation of ≥ 2 additional features.
- ▶ Complexity, creativity, and integration of features.

▶ **Code Quality (Preliminary) (10%)**

- ▶ Code is reasonably organized and understandable.
- ▶ Appropriate use of programming constructs.

Capstone: Analysis, Interpretation and Critical Thinking (30% of Capstone Grade)

- ▶ **Results Demonstration and Visualization (10%)**
 - ▶ Clear and effective presentation of simulation results (graphs, animations, etc.).
 - ▶ Results address project objectives and show model behavior.
- ▶ **Analysis of Parameters and Emergent Patterns (10%)**
 - ▶ Insightful discussion on how parameters/initial conditions affect outcomes.
 - ▶ Identification and explanation of emergent behaviors.
- ▶ **Discussion of Limitations and Real-World Comparison (10%)**
 - ▶ Thoughtful consideration of model limitations and assumptions.
 - ▶ Comparison to real-world data, phenomena, or existing research.

Capstone: Presentation and Communication (20% of Capstone Grade)

▶ **Structure and Clarity (10%)**

- ▶ Well-organized, logical, and easy-to-follow presentation.
- ▶ Clear explanation of goals, methods, features, and results.
- ▶ Effective use of visual aids.

▶ **Delivery and Engagement (5%)**

- ▶ Clear, confident speech with appropriate pacing.
- ▶ Engages the audience effectively.

▶ **Adherence to Time (5%)**

- ▶ Adherence to allocated presentation time (e.g., 20-25 min + Q & A).

Capstone: Abstract (10% of Capstone Grade)

- ▶ **Content and Clarity (5%)**

- ▶ Abstract clearly and concisely summarizes the project's purpose, methods, key findings, and significance.

- ▶ **Accuracy and Completeness (5%)**

- ▶ Accurately reflects the work done and main conclusions.
- ▶ Includes all necessary components of a scientific abstract.

Grading Criteria for Code Review

This component assesses the quality, design, and documentation of the Capstone Project code.

Assumed contribution to final grade: **30%** (adjust as needed)

- ▶ Focuses on the submitted codebase for the Capstone Project.
- ▶ May involve a code walk-through or defense session.

Code Review: Code Quality and Design (50% of Code Review Grade)

▶ **Readability and Maintainability (15%)**

- ▶ Clean, well-formatted, and understandable code.
- ▶ Consistent naming conventions and style.
- ▶ Logical organization (functions, classes, modules).

▶ **Modularity and Reusability (15%)**

- ▶ Code broken into manageable, reusable components.
- ▶ Avoidance of monolithic code blocks.

▶ **Efficiency and Performance (10%)**

- ▶ Consideration for efficiency of algorithms/data structures.
- ▶ Avoidance of obvious performance bottlenecks.

▶ **Error Handling and Robustness (10%)**

- ▶ Basic error handling for invalid inputs/conditions.
- ▶ Code is reasonably robust for typical usage.

Code Review: Documentation and Comments (20% of Code Review Grade)

▶ **Inline Comments (10%)**

- ▶ Sufficient and meaningful comments for complex/non-obvious code.
- ▶ Comments aid in understanding the code's logic.

▶ **Function/Module Documentation (10%)**

- ▶ Clear docstrings or header comments for key functions/modules (purpose, parameters, returns).
- ▶ README or equivalent for setup and execution.

Code Review: Correctness and Functionality (20% of Code Review Grade)

▶ **Adherence to Project Specification (10%)**

- ▶ Code correctly implements core requirements and chosen additional features of the Capstone Project.

▶ **Logical Soundness (10%)**

- ▶ Logic within the code is sound and correctly translates model rules.
- ▶ Absence of significant logical flaws or bugs.

Code Review: Discussion and Response to Review (10% of Code Review Grade)

(Applicable if an interactive review session is held)

- ▶ **Explanation of Choices (5%)**

- ▶ Ability to clearly articulate design decisions and justify coding choices made during development.

- ▶ **Responsiveness to Feedback (5%)**

- ▶ Openness to constructive criticism.
- ▶ Ability to discuss potential improvements or alternatives thoughtfully.